GOVERNMENT EXAM MASTER GUIDE

SSC CGL • Railway NTPC / Group D • Banking • Supreme Court JCA-cum-Junior Programmer

A structured foundation plan, subject notes, formulas, technical basics, practice MCQs and answer explanations

Designed for beginners

Use this as a study workbook: learn a topic, solve questions, review mistakes, then revise.

Important: No PDF can guarantee selection. This guide helps you organise preparation; exact eligibility, syllabus, pattern, dates and skill tests must be checked in the latest official notification for each exam.

1. Your Strategy: Prepare Common Subjects First

You have named four exam groups. Their common subjects are Quantitative Aptitude, Reasoning, English and General Awareness. Banking often requires faster timed reasoning/data interpretation and banking awareness. Railway exams include broad General Awareness and may require a typing/aptitude/medical stage depending on post. SSC CGL has its own tier and paper structure. Supreme Court Junior Court Assistant-cum-Junior Programmer includes a technical/programming element according to the applicable official notice.

Exam group	Shared preparation	Extra preparation to verify
SSC CGL	Maths, Reasoning, English, General Awareness	Tier-wise syllabus, statistics/finance papers for specific posts, DEST/skill require
Railway NTPC / Group D	Maths, Reasoning, General Awareness/Science	Post-wise CBT stages, typing/aptitude/physical/medical criteria
Banking (IBPS/SBI)	Reasoning, Quant, English	Puzzles, DI, banking/economy awareness, descriptive/other stages where notifi
Supreme Court JCA-English, GK, reasoning, quant/computer basics as relevant		Exact technical syllabus, programming language, skill test and recruitment stage

Do not assume one exam's syllabus or pattern applies to another. Keep a separate one-page checklist for each notification.

2. Official Sources — Start Here

• Supreme Court of India recruitment and syllabus: https://www.sci.gov.in/recruitments/

• SSC syllabus and notices: https://ssc.gov.in/for-candidates/syllabus and https://ssc.gov.in/

• Railway Recruitment Board notices: https://www.rrbcdg.gov.in/employment-notices.php (also check the RRB assigned to your application)

• IBPS notifications: https://www.ibps.in/

• SBI Careers: https://sbi.co.in/web/careers

• NCERT textbooks for school-level foundations: https://ncert.nic.in/

Download the detailed advertisement and read: educational qualification, age cut-off, exam stages, number of questions, marks, negative marking, time, typing/skill requirements, documents and deadlines.

3. 24-Week Preparation Roadmap

Phase	Weeks	Work
Foundation	1–4	Arithmetic basics, grammar, reasoning basics, school-level science, polity/history basics; start computer basics
Build concepts	5–8	Percentages, ratio, averages, profit-loss, SI/CI, work/time, speed/distance, series, coding, syllogism
Coverage	9–12	Geometry/mensuration/algebra as needed, comprehension, vocabulary, geography/economy, computer and technic
Exam-specific	13–16	Study each target exam's exact pattern; solve its previous-year papers and sectional tests
Timed practice	17–20	2–4 timed tests per week; analyse every error; practise typing/programming/skill tasks if required
Revision	21–24	Revise notes, repeat weak topics, full mocks under exam conditions, confirm documents and official updates

If your exam is sooner, compress the plan but do not skip the official syllabus, previous-year papers, mock analysis and revision.

4. Suggested Daily Timetable

Time block	Task	How to study
45 min	Quantitative Aptitude	One concept + 15–25 questions; write the shortest valid method
40 min	Reasoning	One topic + timed set; diagram puzzles instead of guessing
30 min	English	Grammar/vocabulary plus one short reading passage
35 min	GK / General Studies	Static topic + short current-affairs notes; revise older notes
45 min	Computer/programming or exam-specific topic	Theory + hands-on coding / MCQs, according to SCI syllabus
15 min	Error notebook	Record why each wrong answer happened and redo it later

For 2 hours daily, shorten each block but keep regular practice and error review. One weekly rest/light revision session can help maintain consistency.

5. Quantitative Aptitude — Formula Sheet

Percentages, ratio and average

Percentage = (part ÷ whole) × 100. x% of y = xy/100. Increase by x%: multiply by (1+x/100). Decrease by x%: multiply by (1−x/100). Ratio a:b with total T gives first share aT/(a+b). Average = sum/count; sum = average × count.

Profit, loss and discount

Profit = SP−CP; Loss = CP−SP. Profit% = Profit/CP × 100. Loss% = Loss/CP × 100. Discount = Marked Price−Selling Price. Discount% = Discount/Marked Price × 100.

Interest

SI = P×R×T/100. Amount under simple interest = P+SI. Annual compound amount A=P(1+R/100)^T; CI=A−P. Match compounding frequency to the question.

Time, speed, distance and work

Speed=Distance/Time. km/h to m/s: ×5/18; m/s to km/h: ×18/5. If A finishes in x days, one-day work=1/x. Combined work rates add when people work independently at constant rates.

Algebra, geometry and data

$(a+b)^2=a^2+2ab+b^2$; $(a-b)^2=a^2-2ab+b^2$; $a^2-b^2=(a-b)(a+b)$. Triangle area=½×base×height; rectangle area=l×w; circle area=πr^2; circumference=$2\pi r$; cuboid volume=lwh; cylinder volume=$\pi r^2 h$. Mean=sum/count. Probability=number of favourable outcomes/total equally likely outcomes.

Fast number facts

Divisibility: 2 (even last digit); 3 and 9 (digit sum); 5 (last digit 0 or 5); 10 (last digit 0). Learn squares 1–30, cubes 1–15 and common fraction-percentage equivalents: 1/2=50%, 1/3≈33.33%, 1/4=25%, 1/5=20%, 1/8=12.5%.

6. Reasoning — Topic Checklist and Method

• Series: test differences, ratios, alternating patterns, squares/cubes, and letter positions.

• Analogy/classification: identify the exact relation shared by the relevant pair or group.

• Coding-decoding: test alphabet shifts, reverse order, positions A=1 to Z=26, and alternating rules.

• Direction sense: draw a compass and track each move; calculate net horizontal and vertical distance.

• Blood relations: build a family tree; never infer unstated relationships.

• Syllogisms: use only statements given. 'All A are B' does not mean 'All B are A'.

• Venn diagrams: draw overlapping circles according to 'all', 'some' and 'none'.

• Seating/puzzles: write fixed facts first, mark exclusions, and do not make unsupported assumptions.

• Statement/conclusion: distinguish what must follow from what might be true.

• Banking focus: timed seating arrangements, inequalities, input-output, data sufficiency and data interpretation as per current pattern.

7. English — Revision Notes

Grammar

• Subject-verb: He works; they work; everyone is ready.

• Articles depend on sound: a university; an hour.

• Tenses: write–wrote–written; go–went–gone; see–saw–seen.

• Common preposition pairs: interested in, good at, depend on, responsible for.

• Revise subject-verb agreement, pronouns, modifiers, parallelism, conjunctions, voice, narration and sentence correction.

Vocabulary and comprehension

Learn 5–10 words daily with meaning, synonym, antonym and an example sentence. Read one short passage daily. Identify main idea, tone, factual detail and inference; answer from the passage.

Practice habits

Keep an error list for spelling, articles, tense, prepositions and confusing words. Redo incorrect questions after one day and again after one week.

8. General Awareness / General Studies

Polity

Revise the Preamble; Fundamental Rights (Part III); Directive Principles (Part IV); Fundamental Duties (Part IVA); Parliament; President; Prime Minister; Supreme Court and High Courts; constitutional bodies; elections; local government. Learn article numbers only from a reliable, current source.

History

Ancient: Indus Valley civilisation, Vedic period, Buddhism/Jainism, Maurya and Gupta. Medieval: major kingdoms, Delhi Sultanate, Mughal period, Bhakti/Sufi traditions. Modern: European companies, British expansion, 1857, reform movements, national movement and independence.

Geography

Latitude/longitude, Earth's motions, atmosphere, climate/monsoon, Indian physical divisions, rivers, soils, crops, minerals, states/capitals, neighbouring countries, important national parks and map practice.

Economy

GDP, inflation, fiscal versus monetary policy, RBI, repo rate, budget, taxation, unemployment, poverty, banking basics and government schemes. Understand definitions before memorising figures.

Science

Physics: motion, force, work, energy, heat, light, sound, electricity. Chemistry: elements, compounds, acids/bases/salts, metals/non-metals. Biology: cell, nutrition, respiration, circulation, immunity, plants and ecology. Railway exams commonly draw on school-level general science; verify your exam notice.

Current affairs

Current affairs become outdated quickly. Make weekly notes on national/international news, economy, government programmes, science, awards, appointments, sports, reports and important days. Verify facts and dates before the exam.

9. Computer Fundamentals

• Hardware is physical equipment; software is instructions/programs.

• CPU executes instructions; RAM is volatile working memory; SSD/HDD are persistent storage.

• An operating system manages files, memory, processes, devices and user interaction.

• Input devices include keyboard/mouse; output devices include monitor/printer.

• LAN covers a limited local area; WAN covers a broader geographic area. A router forwards traffic between networks.

• HTTPS protects data in transit. Use unique passwords, multi-factor authentication, updates and backups.

• Office tools: practise document formatting, spreadsheets, formulas, sorting/filtering, presentations and common keyboard shortcuts if relevant to the notice.

10. Programming Fundamentals — SCI Junior Court Assistant-cum-Junior Programmer

Important: match every topic to the exact SCI syllabus/advertisement for the recruitment you are applying to. Do not assume this checklist is the official technical syllabus.

Programming logic

• Algorithm: finite, ordered steps to solve a problem. Flowchart: visual representation of steps and decisions.

• Variable stores a value; data type defines the kind of value; operator performs an operation.

• Conditionals (if/else) select a path; loops repeat operations; functions encapsulate reusable logic.

• Array/list stores a collection; string stores text; recursion is a function calling itself.

• Debugging sequence: reproduce, isolate, inspect inputs/state, fix, test edge cases, add regression test.

Object-oriented programming

Encapsulation groups data and methods while controlling access; inheritance derives a class from another; polymorphism allows one interface to have different implementations; abstraction exposes essential behaviour while hiding details.

Data structures and algorithms

Understand arrays, linked lists, stacks (LIFO), queues (FIFO), trees, hash tables, searching and sorting at the level specified. Know that linear search is O(n); binary search is O(log n) on sorted data; common comparison sorts often run O(n log n) on average.

Database and SQL basics

Table contains rows and columns; primary key uniquely identifies a row; foreign key relates tables. SELECT reads data; INSERT adds rows; UPDATE changes rows; DELETE removes rows. Always inspect WHERE clauses before UPDATE/DELETE.

Web/network/security basics

Know client-server concepts, HTTP/HTTPS, IP address, DNS, authentication versus authorization, least privilege, backups, common malware/phishing risks and safe handling of data, if included in the official syllabus.

Hands-on practice

Write small programs for even/odd, maximum of three numbers, factorial, prime check, reverse string, palindrome, array maximum, linear/binary search and sorting. Use the language named by the official notice; do not spread effort across several languages without a reason.

11. Exam-Specific Playbooks

SSC CGL

• Use the latest SSC notice for Tier-I/Tier-II structure, subject weightage, negative marking and qualifying skill modules.

• Prioritise arithmetic, algebra, geometry/mensuration, trigonometry/statistics only where applicable, reasoning, English and General Awareness.

• Practise previous-year papers in timed conditions; analyse accuracy by topic.

Railway NTPC / Group D

• Use the relevant CEN and your RRB website; stages and criteria vary by post and recruitment notice.

• Practise maths, reasoning, General Awareness, current events and school-level science.

• Check whether your post requires typing, aptitude, physical efficiency, medical standards or document verification.

Banking (IBPS/SBI)

• Use the current notification for preliminary/main exam structure and section timings.

• Train for speed in arithmetic, data interpretation, puzzles, inequalities, syllogisms and reading comprehension.

• Maintain current banking/economy notes; practise descriptive writing if required by the exam.

Supreme Court JCA-cum-Junior Programmer

• Read the exact official advertisement and syllabus on the SCI recruitment page.

• Prepare the listed general subjects plus the technical/programming topics specifically required.

• Check eligibility, qualification, age cut-off, exam stages, computer/typing/technical skill requirements and notice dates.

• Do not confuse this post with Junior Court Assistant or Junior Court Attendant; titles and requirements differ.

12. Practice Set — 30 Questions

1. 15% of 400 = ?
A 40 B 50 C 60 D 80

2. A ratio is 3:5 and total is 64. The smaller share is:
A 18 B 24 C 32 D 40

3. Average of 12, 16, 20 is:
A 14 B 16 C 18 D 20

4. CP ■800, SP ■920. Profit percent:
A 10% B 12% C 15% D 20%

5. 72 km/h equals:
A 10 m/s B 15 m/s C 20 m/s D 25 m/s

6. A does a job in 8 days. One-day work is:
A 1/4 B 1/6 C 1/8 D 8

7. Series: 4, 9, 16, 25, ?
A 30 B 32 C 36 D 49

8. Series: 3, 6, 12, 24, ?
A 30 B 36 C 42 D 48

9. If A=1, C=3, then CAB = ?
A 3-1-2 B 2-1-3 C 3-2-1 D 1-3-2

10. Opposite direction of South-West:
A NE B NW C SE D SW

11. Correct sentence:
A They is here B They are here C They was here D They be here

12. Correct article: '___ honest person'
A A B An C The only D No article

13. Past tense of 'teach':
A teached B taught C teaching D teaches

14. Synonym of 'brief':
A short B huge C slow D weak

15. Antonym of 'expand':
A grow B enlarge C contract D extend

16. Fundamental Rights are in:
A Part II B Part III C Part IV D Part IVA

17. India's central bank is:
A SBI B RBI C SEBI D NABARD

18. Inflation means:
A general sustained price rise B only population rise C rainfall rise D all prices fall

19. Plants absorb which gas for photosynthesis?
A Oxygen B Carbon dioxide C Helium D Hydrogen

20. SI unit of force:
A Joule B Watt C Newton D Pascal

21. H2O is the formula for:
A salt B water C oxygen D carbon dioxide

22. RAM is generally:
A volatile B permanent-only storage C a printer D an operating system

23. Which is an operating system?
A Windows B Keyboard C Router D HTML

24. Which loop repeats a block of code?
A for/while B comment C variable name D file extension

25. Which SQL command retrieves rows?
A SELECT B DELETE C DROP D UPDATE

26. Stack follows:
A FIFO B LIFO C random only D alphabetical order

27. A primary key should:
A uniquely identify a row B be duplicated in every row C always be a password D delete a table

28. Binary search requires data to be:
A sorted B encrypted C printed D duplicated

29. HTTPS mainly helps protect:
A data in transit B keyboard hardware C battery life D screen brightness

30. Best response to an unknown suspicious email link:
A click quickly B share password C verify sender and avoid opening if suspicious D disable updates

13. Answer Key with Explanations

Q	Answer	Explanation
1	C	400 × 15/100 = 60.
2	B	8 total parts; one part=64/8=8; smaller share=3×8=24.
3	B	48/3=16.
4	C	Profit=120; 120/800×100=15%.
5	C	72×5/18=20 m/s.
6	C	One-day work=1/8.
7	C	Squares 2^2, 3^2, 4^2, 5^2; next 6^2=36.
8	D	Terms double; next 48.
9	A	C=3, A=1, B=2.
10	A	Opposite of SW is NE.
11	B	Plural subject 'they' takes 'are'.
12	B	Honest starts with a vowel sound.
13	B	Teach–taught–taught.
14	A	Brief means short.
15	C	Contract is the opposite of expand.
16	B	Fundamental Rights are in Part III.
17	B	RBI is India's central bank.
18	A	Inflation is a sustained increase in the general price level.
19	B	Plants use carbon dioxide in photosynthesis.
20	C	Newton is the SI unit of force.
21	B	H■O is water.
22	A	RAM is volatile working memory.
23	A	Windows is an operating system.
24	A	for/while loops repeat code.
25	A	SELECT retrieves data.
26	B	Stack is last-in, first-out.
27	A	Primary key uniquely identifies a row.
28	A	Binary search assumes sorted data.
29	A	HTTPS encrypts the connection in transit.
30	C	Verify suspicious messages and avoid unsafe links.

14. Mock Test Analysis Sheet

Metric	Record after each mock
Score / accuracy	Correct __ / attempted __; accuracy = correct ÷ attempted × 100
Time management	Questions left; sections that consumed too much time
Concept errors	Topics you did not understand

Metric	Record after each mock
Careless errors	Calculation, reading, marking or unit mistakes
Next actions	Choose 3 weak topics and redo them within 48 hours

Do not only collect mock scores. Improvement comes from diagnosing and correcting repeated errors.

15. Final 7-Day Revision Checklist

• Day 1: formula sheet, percentages, ratio, averages, profit-loss.

• Day 2: work/time, speed/distance, algebra/geometry as required.

• Day 3: reasoning series, coding, directions, syllogisms, puzzles.

• Day 4: grammar rules, vocabulary notebook, reading comprehension.

• Day 5: polity, history, geography and science short notes.

• Day 6: computer/programming syllabus and one timed technical set for SCI.

• Day 7: one full mock, review mistakes, check admit-card instructions and sleep properly.

16. Personal Selection Plan

Selection depends on eligibility, preparation, accuracy, competition, exam-day performance and the specific recruitment rules. No study guide can guarantee selection. Your best process is: follow official syllabus → learn concepts → practise questions → analyse mistakes → revise → take timed mocks → verify official updates.

Keep one folder per exam with the notification PDF, syllabus checklist, previous-year papers, mock scores, ID/document checklist and revision notes.

Official links: SCI — https://www.sci.gov.in/recruitments/ | SSC — https://ssc.gov.in/ | RRB — https://www.rrbcdg.gov.in/employment-notices.php | IBPS — https://www.ibps.in/ | SBI Careers — https://sbi.co.in/web/careers | NCERT — https://ncert.nic.in/